

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{\begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix}}_{K_c} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{\begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix}}_{D_c} \begin{bmatrix} \dot{p}_d - \dot{p}_{EE} \\ \omega_d - \omega_{EE} \end{bmatrix}$$